

Questions with Answer Keys

#MathBoleTohMathonGo

Q1 - Single Correct

The value of $\sum_{i=0}^n \sum_{j=0}^i \sum_{k=0}^j \sum_{m=0}^k n$ is equal to

(1) $n+1 C_4$

(2) $n \cdot n+1 C_4$

(3) $n+1 C_3$

(4) $n(n+1)$

Q2 - Single Correct

Let $n_1 = x_1 x_2 x_3$ and $n_2 = y_1 y_2 y_3$ be two 3-digit numbers, then the pairs of n_1 and n_2 can be formed so that n_1 can be subtracted from n_2 without borrowing, is

(1) $45 \cdot (55)^2$

(2) $55 \cdot (45)^2$

(3) $55 \cdot (54)$

(4) $45 \cdot (55)$

Q3 - Single Correct

Let $S_n = \sum_{r=1}^n r!$; ($n > 6$). Then, $S_n - 7 \left[\frac{S_n}{7} \right]$ (where, $[\cdot]$ denotes greatest integer function) is equal to

(1) $\left[\frac{n}{7} \right]$

(2) $n! - 7 \left[\frac{n}{7} \right]$

(3) 5

(4) 3

Q4 - Single Correct

A seven-digit numbers divisible by 9 is to be formed by using 7 out of numbers $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. The number of ways in which they can be done, is

(1) $7!$

(2) $2(7)!$

(3) $3(7)!$

(4) $4(7)!$

Q5 - Single Correct

Among $9!$ permutations of the digit $1, 2, 3, \dots, 9$. Consider those arrangements which have the property that, if we take any five consecutive positions, the product of the digits in those positions is divisible by 7, the number of such arrangements is

(1) $7!$

(2) $3 \cdot (7)!$

(3) $8!$

(4) $4(7)!$

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Q6 - Single Correct

The number of ways of arranging m numbers out of $1, 2, 3, \dots, n$ so that maximum is $(n-2)$ and minimum is 2 (repetitions of numbers is allowed) such that maximum and minimum both occur exactly once, ($n > 5, m > 3$) is

- (1) ${}^{n-3}C_{m-2}$
- (2) ${}^mC_2(n-3)^{m-2}$
- (3) $m(m-1)(n-5)^{m-2}$
- (4) ${}^nC_2 \cdot {}^nC_m$

Q7 - Single Correct

The number of rational numbers lying in the interval $(2014, 2015)$ all whose digits after the decimal points are non-zero and are in decreasing order, is

- (1) $\sum_{i=1}^9 {}^9P_i$
- (2) $2^9 - 1$
- (3) $\sum_{i=1}^{10} {}^{10}P_i$
- (4) $2^{10} - 1$

Q8 - Single Correct

The number of ways of dividing 15 men and 15 women into 15 couples, each containing of a man and a woman, is

- (1) 1240
- (2) 1840
- (3) 1820
- (4) 2005

Q9 - Single Correct

' n ' is selected from the set $\{1, 2, 3, \dots, 100\}$ and the number $2^n + 3^n + 5^n$ is formed. Total number of ways of selecting ' n ', so that the formed number is divisible by 4, is equal to

- (1) 50
- (2) 49
- (3) 48
- (4) None of these

Q10 - Single Correct

In a shooting competition a man can score 5, 4, 3, 2 or 0 points for each shot. Then, the number of different ways in which he can score 30 in seven shots, is

- (1) 419
- (2) 418
- (3) 420

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(4) 421

Q11 - Single Correct

The sum of the factors $8!$ which are odd and are the form $3m + 2$, where m is a natural number is

(1) 8

(2) 40

(3) 45

(4) 50

Q12 - Single Correct

The number of divisors of $n = 2^7 \cdot 3^5 \cdot 5^3$ which are of the form $4t + 1$, ($t \in N$), is

(1) 12

(2) 11

(3) 10

(4) 9

Q13 - Single Correct

The number of ways of choosing triplets (x, y, z) such that $z > \max\{x, y\}$ and $x, y, z \in \{1, 2, 3, \dots, (n+1)\}$, is

(1) $n(n+1)(n+2)$

(2) $n(n+1)(2n+1)$

(3) $\frac{n(n-1)(2n-1)}{6}$

(4) ${}^{n+1}C_3 + {}^{n+2}C_3$

Q14 - Single Correct

The integers from 1 to 1000 are written in order around a circle, starting at 1, every fifteenth number is marked $\{1, 16, 31, \dots\}$. This process is continued until a number is reached which has already been marked. Then, unmarked numbers are

(1) 200

(2) 400

(3) 600

(4) 800

Q15 - Single Correct

The number of ways in which we can choose 3 squares on a chess board such that one of the squares has its two sides common to the other two squares, is

(1) 290

(2) 292

(3) 294

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(4) 296

Q16 - Single Correct

Let $f(n)$ denote the number of different ways the positive integer 'n' can be expressed as the sum of 1's and 2's for example $f(4) = 5$, since $4 = 1 + 1 + 1 + 1, 1 + 1 + 2, 1 + 2 + 1, 2 + 1 + 1, 2 + 2$. (Note That order of 1's and 2's is important) Then, $f(f(6))$ is

(1) $f(6)$

(2) $f(12)$

(3) $f(13)$

(4) None of these

Q17 - Single Correct

A seven-digit numbers made up of all distinct digits 8, 7, 6, 4, 2, x and y is divisible by 3. Then, possible number of ordered pair (x, y) is

(1) 4

(2) 8

(3) 2

(4) None of these

Q18 - Single Correct

A delegation of 5 students is to be formed from a group of 10 students. If three particular students want to remain together and two particular students do not want to remain together, then number of selections, is

(1) 10

(2) 20

(3) 30

(4) 40

Q19 - Single Correct

Let $a, b, c \in I^+$ such that LCM of a and b is 1000, LCM of b and c is 2000 and LCM of c and a is 2000, then number of ordered triplet (a, b, c) is

(1) 63

(2) 90

(3) 70

(4) None of these

Q20 - Single Correct

The number of ways to place 11 identical balls in three boxes, so that any two boxes together will contain more balls than the other one

(1) 21

(2) 63

(3) 15

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(4) None of these

Q21 - Multiple Correct

Number of ways in which 3 numbers in AP can be selected from $1, 2, 3, \dots, n$ is

- (1) $\left(\frac{n-1}{2}\right)^2$, if n is even
- (2) $\frac{n(n-2)}{4}$, if n is odd
- (3) $\frac{(n-1)^2}{4}$, if n is odd
- (4) $\frac{n(n-2)}{4}$, if n is even

Q22 - Multiple Correct

The combinatorial coefficient ${}^{n-1}C_p$ denotes

- (1) The number of ways in which n things of which p are alike and rest different can be arranged in a circle
- (2) The number of ways in which p different things can be selected out of n different thing, if a particular thing is always excluded
- (3) Number of ways in which n alike balls can be distributed in p different boxes, so that no box remains empty and each box can hold any number of balls.
- (4) The number of ways in which $(n-2)$ white balls and p black balls can be arranged in a line. If black balls are separated, balls are all alike except for the colour.

Q23 - Multiple Correct

The number of ways in which five different books to be distributed among 3 persons, so that each person gets atleast one book, is equal to the number of ways in which

- (1) 5 persons are allotted 3 different residential flats, so that each person is allotted at most one flat and no two persons are allotted the same flat.
- (2) Number of parallelograms formed by one set of 6 parallel lines and other set of 5 parallel lines that goes in other direction.
- (3) 5 different toys are to be distributed among 3 children, so that each child get atleast one toy.
- (4) 3 mathematics professors are assigned five different lectures to be delivered. so that each professor gets atleast one lecturer.

Q24 - Matrix Match (Only one option is correct)

Column I		Column II	
i.	Four different movies are running in a town. Ten students go to watch these four movies. The number of ways in which every movie is watched by atleast one student, is (Assume each way differs only by number of students watching a movie).	p.	11
ii.	Consider 8 vertices of a regular octagon and its centre. If T denotes the number of triangles and S denotes the number of straight lines that can be formed with these 9 points, then the value of $(T - S)$ is	q.	36
iii.	In an examination, 5 children were found to have their mobiles in their pocket. The invigilator fixed them and took their mobiles in his possession. Towards the end of the test, invigilator randomly returned their mobiles. The number of ways in which atleast two children did not get their own mobiles, is	r.	52
iv.	The product of the digits of 3214 is 24. The number of 4 digit natural number such that the product of their digits is 12, is	s.	60
v.	The number of ways in which a mixed double tennis game can be arranged from amongst 5 married couple, if no husband and wife plays in the same game, is	t.	84

(1)	(i)	(ii)	(iii)	(iv)	(v)
	p	q	r	s	t
(2)	(i)	(ii)	(iii)	(iv)	(v)
	q	r	s	t	p
(3)	(i)	(ii)	(iii)	(iv)	(v)
	t	r	p	q	s
(4)	(i)	(ii)	(iii)	(iv)	(v)
	r	s	t	p	q

Q25 - Paragraph

Passage I (For Question 25, 26)

Let a_n denotes the number of all n -digit positive integers formed by the digits 0,1 or both such that no consecutive digit in them are 0. Let b_n = The number of such n -digit integers ending with digit 1 and C_n = the number of such n -digit integers ending with digit 0.

The value of b_6 , is

- (1) 7
(2) 8
(3) 9
(4) 11

Q26 - Paragraph

Which of the following is correct?

- (1) $a_{17} = a_{16} + a_{15}$
(2) $c_{17} \neq c_{16} + c_{15}$
(3) $b_{17} \neq b_{16} + c_{16}$
(4) $a_{17} = c_{17} + b_{16}$

Q27 - Paragraph

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Passage II (For Question 27, 28, 29)

10-digit numbers are formed by using all the digits $0, 1, 2, \dots, 9$ such that they are divisible by 11111. Then, answer the following questions.

The digit in tens place, in smallest of such number, is

(1) 9

(2) 8

(3) 7

(4) 6

Q28 - Paragraph

The digit in units place, in greatest of such numbers, is

(1) 4

(2) 3

(3) 2

(4) 1

Q29 - Paragraph

The total number of such numbers, is

(1) 6543

(2) 5634

(3) 3456

(4) 4365

Q30 - Paragraph

Passage III (For Question 30, 31, 32)

A shopkeeper has unlimited red, green, white, yellow and black balls. All balls of same colour are identical. Then,

In how many ways a person can make a selection of 12 balls

(1) ${}^{15}C_3$

(2) ${}^{16}C_4$

(3) ${}^{16}C_5$

(4) None of these

Q31 - Paragraph

In how many ways a person can select 12 balls, if each selection consists of balls of all colours

(1) ${}^{12}C_3$

(2) ${}^{11}C_4$

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(3) ${}^{13}C_4$

(4) ${}^{15}C_4$

Q32 - Paragraph

In how many ways a person can select 16 balls, if each selection consists of atleast two balls of each colour, is

(1) ${}^{10}C_2$

(2) ${}^{11}C_4$

(3) ${}^{10}C_4$

(4) None of these

Q33 - Paragraph

Passage IV (For Question 33, 34, 35)

Let $x_1x_2x_3x_4x_5x_6$ be a six-digit numbers, find the number of such numbers such that

$x_1 < x_2 < x_3 < x_4 < x_5 < x_6$, is

(1) 9C_6

(2) ${}^{10}C_3$

(3) ${}^{10}C_6$

(4) ${}^{10}C_5$

Q34 - Paragraph

$x_1 < x_2 < x_3 = x_4 < x_5 = x_6$, is

(1) 9C_3

(2) 9C_4

(3) ${}^{10}C_3$

(4) ${}^{10}C_4$

Q35 - Paragraph

$x_1 < x_2 < x_3 \leq x_4 < x_5 < x_6$, is

(1) 9C_4

(2) 9C_3

(3) ${}^{10}C_4$

(4) ${}^{10}C_3$

Q36 - Paragraph

Passage V (For Question 36, 37, 38)

Five balls are to be placed in three boxes. Each can hold all the five balls. In how many different ways can we place the balls so that no box remains empty, if

Questions with Answer Keys

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Balls and boxes are all different

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(1) 100

(2) 125

(3) 150

(4) 50

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Q37 - Paragraph

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Balls are identical but boxes are different

(1) 6

(2) 4

(3) 8

(4) 2

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Q38 - Paragraph

Balls are different but boxes are identical

(1) 15

(2) 10

(3) 35

(4) 25

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Q39 - Paragraph

Passage VI (For Question 39, 40)

A triangle is called an integral triangle, if all its sides are integers. If a, b and c are sides of an integral triangle, then we can assume that $a \leq b \leq c$, because any other permutation of a, b and c will yield the same result. Now, answer the following questions.

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The number of integral isosceles or equilateral triangle's, none of whose side exceeds 4, is

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(1) 11

(2) 12

(3) 9

(4) 10

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Q40 - Paragraph

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The number of integral isosceles or equilateral triangles, none of whose side exceeds $2p$, is

(1) $4p^2 - 2p$

(2) $2p^2 + 2p$

(3) $3p^2$

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Questions with Answer Keys

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(4) $\frac{3p^2}{2} + 3p$

Q41 - Integer Type

Let P_n denotes the number of ways in which three people can be selected out of 'n' people sitting in a row, if no two of them are consecutive. If, $P_{n+1} - P_n = 15$, then the value of n , is

Q42 - Integer Type

Number of cyphers at the end of $^{2002}C_{1001}$, is

Q43 - Integer Type

Given 11 points, in which 5 lie on one circle, other than these 5, no 4 lie on one circle. Then, the maximum number of circle that can be drawn, so that each contains atleast three of the given points, is $52 \cdot (k)$. Then, k is

Q44 - Integer Type

There are 2 identical white balls, 3 identical red balls and 4 green balls of different shades. The number of ways in which they can be arranged in a row so that atleast one ball is separated from the balls of same colour is $K(7! - 4!)$, then k is

Q45 - Integer Type

There are four balls of different colours and four boxes of colours, same as those of the balls. The number of ways in which the balls, one each in a box could be placed such that a ball does not go to a box of its own colour, is

Q46 - Integer Type

There are 100 different books in a shelf. Number of ways in which 3 books can be selected so that no two of which are neighbours is $^{98}C_k$, then k is

Q47 - Integer Type

Number of permutations of 1,2,3,4,5,6,7,8 and 9 taken all at a time, such that the digit

1 appearing somewhere to the left of 2,

3 appearing to the left of 4 and

5 somewhere to the left of 6 is $k \cdot (7!)$, then k is

Q48 - Integer Type

Number of different natural numbers which are smaller than two hundred million and using only the digits 1 or 2 is $k(2^8) - 2$, then k is

Q49 - Integer Type

5 Indian and 5 American couples meet at party and shake hand. If no wife shake hand with her own husband and no Indian wife shakes hands with a male, then the number of handshakes that take place in the party is $15 \cdot k$, then k is

Q50 - Integer Type

Questions with Answer Keys

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Answer Key

Q1 (2)	Q2 (1)	Q3 (3)	Q4 (4)
Q5 (3)	Q6 (3)	Q7 (2)	Q8 (1)
Q9 (2)	Q10 (3)	Q11 (2)	Q12 (2)
Q13 (4)	Q14 (4)	Q15 (2)	Q16 (3)
Q17 (2)	Q18 (2)	Q19 (3)	Q20 (3)
Q21 (3,4)	Q22 (2,4)	Q23 (2,3,4)	Q24 (3)
Q25 (2)	Q26 (1)	Q27 (4)	Q28 (1)
Q29 (3)	Q30 (2)	Q31 (2)	Q32 (3)
Q33 (1)	Q34 (2)	Q35 (3)	Q36 (4)
Q37 (2)	Q38 (1)	Q39 (2)	Q40 (3)
Q41 (8)	Q42 (1)	Q43 (3)	Q44 (6)
Q45 (9)	Q46 (3)	Q47 (9)	Q48 (3)
Q49 (9)	Q50 (8)		